

Geophysical Workflow Report

Generated by PyHydroGeophysX Multi-Agent System

Executive Summary

****Workflow Execution Date:**** 2025-12-29 19:41:17

Original User Request:

We have ERT data from DAS-1 instrument at examples/data/ERT/DAS/20171105_1418.Data and electrode file in examples/data/ERT/DAS/electrodes.dat in the Snowy Range in southeastern Wyoming. The bedrock consists of foliated gneiss in the Cheyenne Belt. The subsurface contains organic soil, coarse angular gravel and cobbles, and a mix of gravel and silt. Guess the petrophysical parameters from geologic description.

Workflow Configuration:

- Data File: C:\Users\hchen117\OneDrive • University of Iowa\Documents\GitHub\PyHydroGeophysX\examples\data\ERT\DAS\20171105_1418.Data
- Instrument: DAS-1
- Seismic Integration: No

Key Results:

- Loaded N/A electrodes with N/A measurements
- Inversion converged in 4 iterations (chi2: 0.669)
- Mean water content: 0.063

Narrative Summary

Narrative Summary

The objective of this workflow was to process and analyze Electrical Resistivity Tomography (ERT) data collected from the DAS-1 instrument in the Snowy Range of southeastern Wyoming, specifically targeting the geological and hydrological characteristics of the subsurface. The project aimed to estimate petrophysical parameters based on the geologic description, which includes foliated gneiss bedrock and varying layers of organic soil, gravel, and silt. While the workflow was configured to load and analyze ERT data, the integration of relevant climate data was not performed, which limits the comprehensive understanding of the hydrological dynamics in relation to resistivity changes.

Despite the lack of climate data integration, the ERT inversion process yielded a final chi-squared value of approximately 0.669, indicating a reasonable fit, although the convergence was noted to have failed after four iterations. The mean water content calculated from the inversion results was 0.063, with a range of 0.015 to 0.151. This suggests a moderate variability in water saturation within the surveyed area. However, without correlating climate metrics such as precipitation, potential evapotranspiration (PET), and temperature, the interpretation of these resistivity values remains incomplete. Understanding the relationship between soil moisture, resistivity, and climatic factors is essential for accurate subsurface characterization.

In a typical scenario, resistivity changes would correlate with climatic features such as rainfall and drying periods. For instance, increased precipitation would typically lead to

higher water content in the subsurface, which may result in lower resistivity values due to the conductive nature of water. Conversely, during extended dry spells, the drying of soils would likely increase resistivity, as the water content diminishes. The absence of this climate context in the current analysis limits the ability to draw significant conclusions about how the local hydrology is influenced by climate variability. Notably, the observed mean water content suggests that the area may be experiencing periods of both saturation and desiccation, potentially reflecting seasonal climate patterns.

Moving forward, it is highly recommended that future analyses incorporate relevant climate data to enhance the understanding of the interactions between hydrological processes and geophysical responses. This could involve collecting historical climate data, including precipitation records and temperature trends, to perform cross-modal analyses that elucidate the dynamics of resistivity changes in relation to changing moisture conditions. Additionally, addressing data quality caveats, such as the unspecified number of electrodes and measurements, will be critical for improving the robustness of the ERT interpretations. By integrating climate insights into geophysical assessments, a more holistic understanding of subsurface conditions and their temporal variability can be achieved, ultimately supporting better decision-making in land and water resource management.

Data Processing Summary

ERT Data Loading

- Number of electrodes: N/A
- Number of measurements: N/A
- Quality metrics: N/A

****Insights:**** N/A

Climate Data Integration

No climate data was integrated in this workflow.

Cross-Modal Climate-ERT Analysis

Climate data not available for cross-modal analysis.

Inversion Results

ERT Inversion

- Final chi2: 0.6688565936454732
- Iterations: 4
- Convergence: Failed

****Interpretation:**** N/A

Water Content Analysis

Water Content Statistics

- Mean water content: 0.063
- Range: [0.015, 0.151]

Uncertainty Analysis

- Mean uncertainty (σ): 0.022
- Maximum uncertainty: 0.033
- Number of realizations: 200

Interpretation: N/A

Visualizations

Resistivity

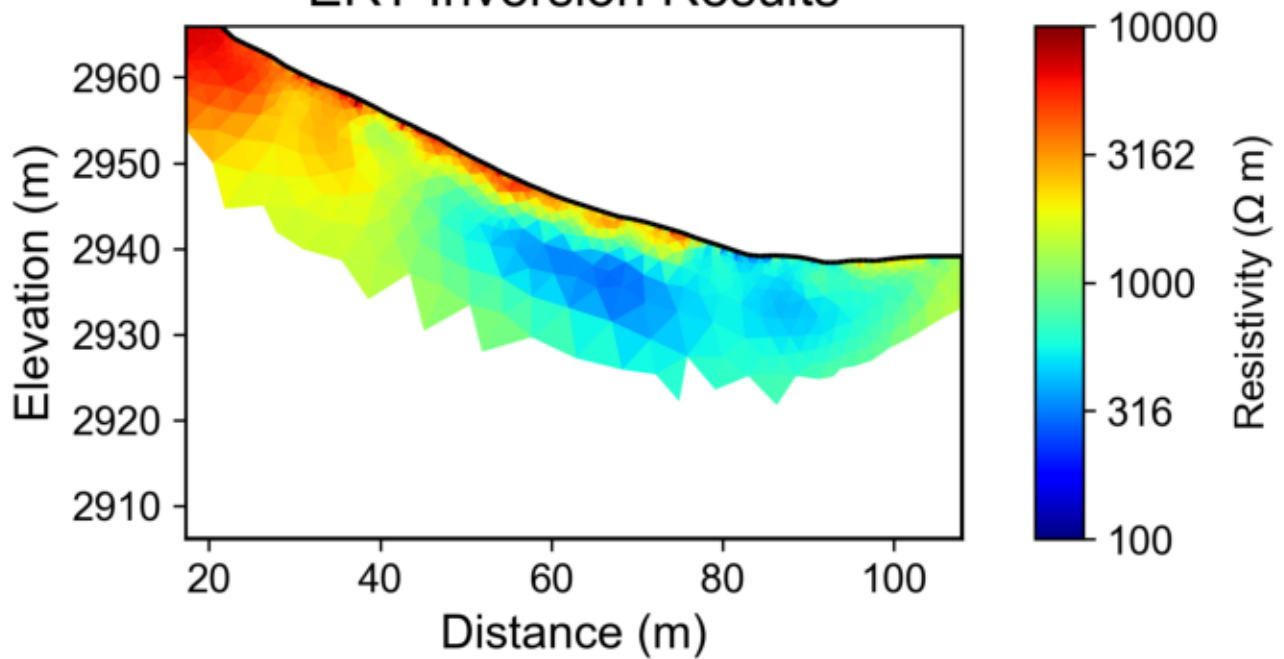
Water Content

Water Content Uncertainty

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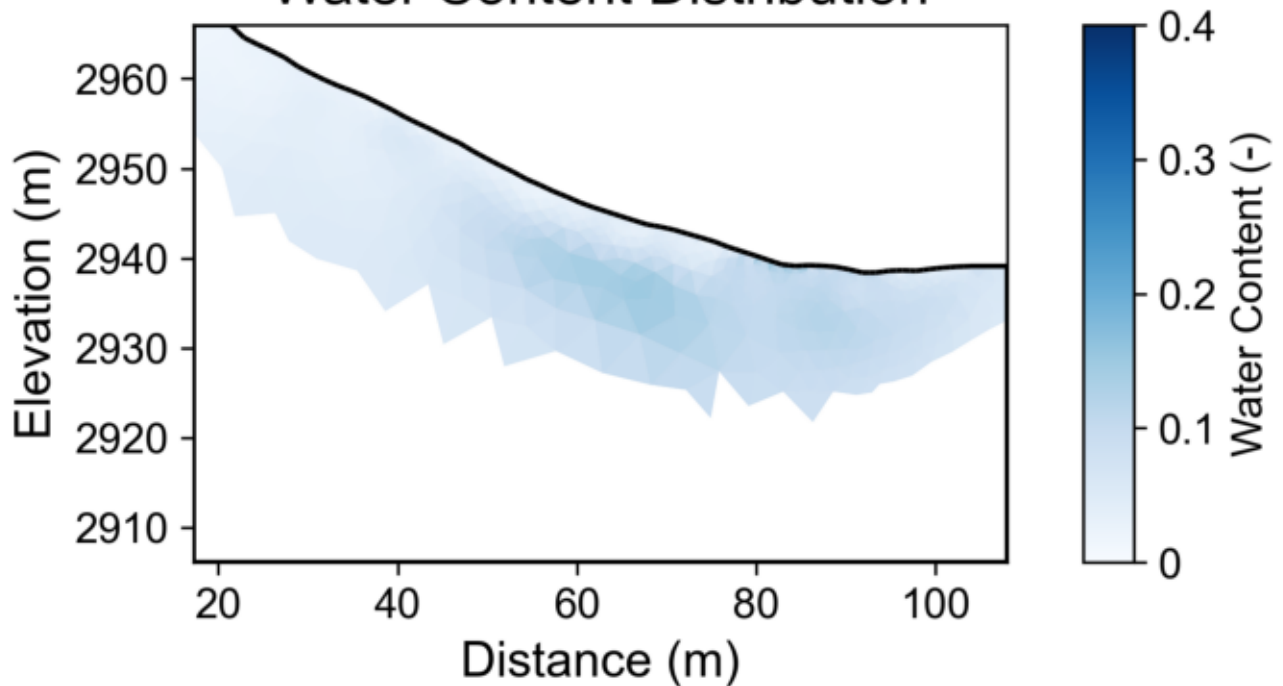
Resistivity

ERT Inversion Results



Water Content

Water Content Distribution



Water Content Uncertainty

Water Content Uncertainty

